

Article

Insights into Boron-based Polyanion-tuned High-nickel Cathodes for High-energy-density Lithium-ion Batteries

Qiang Xie, Wangda Li, Andrei Dolocan, and Arumugam Manthiram

Chem. Mater., **Just Accepted Manuscript** • DOI: 10.1021/acs.chemmater.9b02916 • Publication Date (Web): 03 Oct 2019

Downloaded from pubs.acs.org on October 6, 2019

Just Accepted

"Just Accepted" manuscripts have been peer-reviewed and accepted for publication. They are posted online prior to technical editing, formatting for publication and author proofing. The American Chemical Society provides "Just Accepted" as a service to the research community to expedite the dissemination of scientific material as soon as possible after acceptance. "Just Accepted" manuscripts appear in full in PDF format accompanied by an HTML abstract. "Just Accepted" manuscripts have been fully peer reviewed, but should not be considered the official version of record. They are citable by the Digital Object Identifier (DOI®). "Just Accepted" is an optional service offered to authors. Therefore, the "Just Accepted" Web site may not include all articles that will be published in the journal. After a manuscript is technically edited and formatted, it will be removed from the "Just Accepted" Web site and published as an ASAP article. Note that technical editing may introduce minor changes to the manuscript text and/or graphics which could affect content, and all legal disclaimers and ethical guidelines that apply to the journal pertain. ACS cannot be held responsible for errors or consequences arising from the use of information contained in these "Just Accepted" manuscripts.

Insights into Boron-based Polyanion-tuned High-nickel Cathodes for High-energy-density Lithium-ion Batteries

Qiang Xie, Wangda Li, Andrei Dolocan, and Arumugam Manthiram*

Materials Science and Engineering Program & Texas Materials Institute, The University of Texas at Austin, Austin, TX 78712, USA

ABSTRACT: Ultrahigh-Ni layered oxide cathodes (Ni content > 90%) are at the forefront for potentially enabling higher-energy-density lithium-ion batteries. Unfortunately, they suffer from rapid capacity fade, poor thermal stability, and increased air sensitivity. Introduction of boron-based polyanion into layered cathodes has been found to be effective in stabilizing high-Ni cathodes, but the mechanism behind it is not yet fully understood. To develop a better understanding, we present here an extensive characterization of the ultrahigh-Ni cathode $\text{LiNi}_{0.94}\text{Co}_{0.06}\text{O}_2$ (NC) and $\text{B}_2\text{O}_3\text{-LiNi}_{0.94}\text{Co}_{0.06}\text{O}_2$ (B-NC). The B-NC delivers a capacity of 223 mA h g^{-1} with 80% capacity retention after 400 cycles in full cells with graphite anodes, superior to 61% retention for NC. In-depth interphase investigations reveal that the improved cyclability is attributed to the development of a well-passivated boron/phosphorus-rich cathode-electrolyte interphase in B-NC. Moreover, B-NC demonstrates a considerably enhanced air-exposure and thermal stability. A 30-day air-stored B-NC delivers a discharge capacity of 125 mA h g^{-1} at 10C rate, in sharp contrast to 65 mA h g^{-1} for the stored NC. This work advances the understanding of the effect boron-based polyanion insertion into layered oxides has on battery performance and offers a viable strategy for developing practically viable ultra-high-Ni cathodes with high-energy density.

1. Introduction

The development of Li-ion batteries (LIBs) has enabled ubiquitous access to portable devices with great commercial success. To translate this accomplishments for electric vehicles (EVs) and grid storage, intensive efforts are being devoted to develop advanced cathode materials with higher specific energy at a reduced cost.¹ Ultrahigh-Ni cathodes are an attractive alternative, offering both increased capacity and reduced cost simultaneously.¹⁻³ However, the viability of these materials is hindered by inferior capacity retention, increased reactivity with air, and decreased thermal stability.²⁻⁴ The principal culprits for the capacity fade of high-Ni materials are the parasitic reactions at the cathode-electrolyte interface (CEI) and the anisotropic lattice volume changes during cycling.⁵⁻⁸ The parasitic reactions result in transition-metal dissolution, production of highly-resistive species (such as LiF), and surface structure reconstruction, severely degrading the performance.^{6,9-11} The microcracks generated in the cathode particles upon cycling can exacerbate the interfacial reactions and capacity decay by providing fresh reaction surface.^{4,8} Alleviating these parasitic reactions at the CEI would benefit the long-term cyclability of LIBs.

Chemical instability in air and thermal instability are also challenging issues plaguing the high-Ni cathodes. Exposure of high-Ni cathodes to air triggers complex reactions between air, surface Li^+ , and the unstable lattice oxygen, resulting in a layer of residual lithium (mainly consisting of LiOH , LiHCO_3 , and Li_2CO_3) and NiO on the particle surface.^{1,12-14} The resultant surface residual lithium could deteriorate the performance of high-Ni cathodes, cause potential safety risks related to the generation of CO/CO_2 during cycling, and complicate the suspension

of the cathode slurry due to increased alkalinity.^{1,15,16} In addition, the increased instability of the oxygen lattice structure in the highly delithiated high-Ni cathodes can trigger O_2 release, resulting in catastrophic thermal runaway.¹⁶⁻¹⁸ Stabilizing the oxygen lattice of high-Ni cathodes is critical for improving both the air stability and the thermal stability.

The two principle strategies for solving the stated issues are surface modification and metal-ion doping. Surface modification with metal oxides/fluorides/phosphates can alleviate the electrolyte attack on cathode materials.¹⁹⁻²³ Cationic doping (Al^{3+} , Mg^{2+} , Ti^{4+} , Ga^{3+} *etc.*) can improve the capacity retention by enhancing the stability of the crystal lattice.^{8,24-27} Recently, optimizing layered cathodes by boron-based polyanion (also called boron doping) has attracted increasing attention.^{4,28-32} The incorporation of the boron-based polyanion could stabilize the oxygen close-packed structure of layered oxides by lowering the top of O^{2-} 2p band and introducing stronger B-O bonds (B-O: 809 kJ mol^{-1} , Ni-O: 382 kJ mol^{-1}), which was claimed to improve the cycling performance.²⁸⁻³⁰ Alternatively, Park *et al* proposed that boron doping could improve the cycling performance of high-Ni cathodes by optimizing the secondary-particle microstructure.⁴ This role is distinct from the function of traditional cation-doping/surface coating. While the enhanced cyclability resulting from a small-dose cationic substitution or surface coatings is always at the expense of lower capacity, with a small-dose of boron-based polyanion incorporation, the capacity reduction is limited. The boron-based polyanion has been pointed out in some reports to even improve the capacity and cyclability simultaneously.^{4,28,30-32} All of these merits highlight the potential of boron-based polyanion in optimizing ultrahigh-Ni cathodes for higher-energy-density LIBs.

However, several critical questions remain concerning the boron-based polyanion-tuned layered oxides. First, it has been assumed that all boron from the boron sources (commonly $\text{B}_2\text{O}_3/\text{H}_3\text{BO}_3$) is doped into the host lattice structure (the maximum amount in the previous studies was 8 mol% boron), but compelling evidence was not offered to exclude the possible existence of residual boron sources or other boron-bearing species in the reported materials.^{4,28–32} It is necessary to figure out if residual boron sources/other boron-bearing species are present, and if they are not doped into the host structure, where they are distributed in the material. Second, preceding studies mainly ascribed the enhanced electrochemical performance of the boron-based polyanion-tuned layered cathodes to the improved oxygen packing stability or optimized secondary-particle microstructure. However, in-depth investigation is still required to understand the potential impact of $\text{B}_2\text{O}_3/\text{H}_3\text{BO}_3$ on the cathode-electrolyte interface, as well as its correlation with the enhanced electrochemical performance. Third, systematic studies of the air sensitivity and thermal stability of boron-based polyanion-tuned ultrahigh-Ni cathodes are lacking despite the importance.

With these questions in mind, this study offers an extensive investigation of an ultrahigh-Ni model material $\text{LiNi}_{0.94}\text{Co}_{0.06}\text{O}_2$ with and without B_2O_3 addition (3 mol% boron). With the help of time-of-flight secondary ion mass spectrometry (TOF-SIMS) and X-ray diffraction (XRD) analysis, it is found that certain amount of BO_x^- and LiBO_2^- species are present on the surface and interior of the fresh B_2O_3 - $\text{LiNi}_{0.94}\text{Co}_{0.06}\text{O}_2$ (hereafter termed as B-NC) in addition to the atomic boron doped into the host structure. Electrochemical characterizations show that B-NC delivers a high capacity of 223 mAh g⁻¹ at C/3 with 80% capacity retention after 400 cycles in full cell paired with graphite, greatly outperforming the 61% retention of pure $\text{LiNi}_{0.94}\text{Co}_{0.06}\text{O}_2$ (hereafter termed as NC). Structure and interface analysis indicate that the formation of a compact and well-passivated phosphorus/boron-rich CEI predominantly contributes to the improved cyclability of B-NC. Results from chemical titration, electrochemical measurements, and differential scanning calorimeter imply that B-NC has higher air and thermal stability compared to NC. Overall, this article presents a comprehensive mechanistic study of the effect boron-based polyanion has in stabilizing high-Ni cathodes and confirms the practical relevance of this system in developing advanced LIBs.

2. Experimental Section

Materials Synthesis: The model $\text{Ni}_{0.94}\text{Co}_{0.06}(\text{OH})_2$ precursor was synthesized by a co-precipitation method as we reported before.²⁴ Specific procedures are as follows: A 1.0 M mixed aqueous solution of $\text{NiSO}_4 \cdot 6\text{H}_2\text{O}$ and $\text{CoSO}_4 \cdot 7\text{H}_2\text{O}$ (molar ratio of Ni to Co is 94 : 6) was pumped into a batch tank reactor, accompanied by a continuous feeding of 2.0 M KOH solution and concentrated NH_4OH solution. In the meantime, the pH (10–11), temperature (50 °C), and stirring speed (800 rpm) were well-controlled during the coprecipitation. After washing with deionized water, filtering, and drying at 100 °C overnight, the hydroxide precursor $\text{Ni}_{0.94}\text{Co}_{0.06}(\text{OH})_2$ was mixed with $\text{LiOH} \cdot \text{H}_2\text{O}$ in a Li : (Ni + Co) molar ratio of 1.03 : 1 and calcined at 640 °C for 15 h under oxygen atmosphere to obtain $\text{LiNi}_{0.94}\text{Co}_{0.06}\text{O}_2$. For the B_2O_3 -added $\text{LiNi}_{0.94}\text{Co}_{0.06}\text{O}_2$, $\text{Ni}_{0.94}\text{Co}_{0.06}(\text{OH})_2$ precursor was first homogeneously mixed with B_2O_3 in absolute ethanol (molar ratio of B to (Ni + Co) = 3 : 97) and then dried until the complete evaporation of ethanol. Afterward, the obtained mixture was calcinated with $\text{LiOH} \cdot \text{H}_2\text{O}$ at

680 °C for 15 h under oxygen atmosphere (Li : (Ni + Co + B) = 1.03 : 1).

Electrochemical tests: The cathode electrodes were fabricated by casting the cathode slurries onto an Al foil with an active material loading of ~ 5 mg cm⁻²; the slurries were prepared by mixing the active material, carbon black, and poly(vinylidene fluoride) (PVDF) (weight ratio = 8 : 1 : 1) in N-methyl-2-pyrrolidone (NMP). Half-cell performances were evaluated in 2032 coin cells that were fabricated by assembling the positive electrodes, electrolyte, Celgard 2500 separator, and Li-metal anode (MTI Corp). The electrolyte solution is 1.0 M LiPF_6 in ethylene carbonate (EC)/ethylmethyl carbonate (EMC) (3:7 by weight) with 2 wt.% vinylene carbonate (VC) additive. The assembled coin cells were cycled on an Arbin instrument between 2.8 and 4.4 V at C/3 rate and 25 °C (1C = 180 mA g⁻¹). Long-term cyclability performances were evaluated in pouch-type full cells fabricated with the cathode and a mesocarbon microbead graphite anode with a N/P ratio of ~ 1.08. The electrolyte used in pouch full cells is the same as that used in half cell, namely, EC/EMC (3:7 by weight) with 2 wt.% VC. Pouch-full cells were cycled at C/10 rate (current density 18 mA g⁻¹) for 3 formation cycles and subsequently at C/2 rate for 400 times between 2.5 and 4.3 V at 25 °C with a constant current-constant voltage charging mode. Galvanostatic intermittent titration technique (GITT) measurement on both the samples was carried out on Arbin battery cycler at room temperature. A 600 s galvanostatic charge pulse (0.1C) was applied, followed by an open-circuit rest of 3600 s. The test procedure was maintained until an upper cut-off voltage of 4.4 V.

Characterization: The morphology of both samples was examined with a scanning electron microscope (SEM) (FEI Quanta 650). Cross-sectional samples, which were used for cross-sectional SEM and TOF-SIMS analysis, were prepared by ion-milling (IM4000 plus, Hitachi). X-ray powder diffraction (XRD) was used to study the structure and phase purity of the fresh and stored samples from 10 to 80 ° with a 0.02 ° scan step (Rigaku Miniflex 600, Cu K α radiation). Fullprof program was utilized to perform the Rietveld refinement. *In-situ* XRD test was performed to investigate the phase transition of both samples upon charging.

TOF-SIMS analysis was carried out with a TOF-SIMS 5 spectrometer (ION-TOF GmbH). An ultrahigh vacuum environment was maintained in the analysis chamber with an internal pressure of < 2 × 10⁻⁹ mbar. All the specific detected secondary ions, with negative/positive polarity, had a mass resolution of > 5000. A pulsed 30 keV Bi_1^+ (20 ns) ion beam set was applied for depth profiling or high lateral resolution mapping (< 200 nm), respectively, in the high current mode or burst alignment mode. A 500 eV Cs^+ (negative) or a 1 keV O_2^+ (positive) ion beam, depending on the secondary ion polarity, was used to sputter the cycled composite electrodes. The typical sputtering and analysis areas were, respectively, 300 μm × 300 μm and 100 μm × 100 μm .⁹ The sputtering rate was measured at ~ 0.04 nm s⁻¹ for Cs^+ . XPS study of the cycled samples was performed on an Axis Ultra DLD spectrometer (Kratos) with Al K α radiation source (1486.6 eV). The cycled samples for TOF-SIMS and XPS analysis were rinsed with dimethyl carbonate, dried in an argon-filled glovebox, and then transferred to the instruments with a novel air-free interface.

The residual lithium content of each sample was determined by a chemical titration. Briefly, 3 g of sample was immersed

into 30 mL of deionized water and then sonicated for 5 min to dissolve residual surface lithium. Afterward, the mixture was filtered to collect a clear solution, which was then titrated with 0.1 M HCl solution (Mettler Toledo Titrator). The consumed HCl at the titration endpoint was used to calculate the residual lithium content of each sample. To compare the air sensitivity of both the cathodes, the residual lithium content of both the fresh and 30-day stored samples were tested. The stored samples were obtained by storing the freshly synthesized materials (sealed in glass vial) in air for 30 days.

Thermal stability was evaluated by differential scanning calorimeter (DSC) (Netzsch STA 449). For DSC tests, both samples were first cycled 3 times in a coin cell (vs Li/Li⁺) at C/5 rate from 2.8 to 4.4 V and subsequently charged to 4.3 V at C/10 rate. The charged samples were disassembled and washed with dimethyl carbonate (DMC) in an argon-filled glovebox. Afterward, the charged samples were collected and sealed in a stainless crucible with ~35 wt.% electrolyte added. DSC measurement was performed at a heating rate of 1 °C min⁻¹ from 30 to 350 °C under Ar atmosphere.

3. Results and Discussion

3.1 Distribution Profile of Boron-related Species

The secondary particle morphology in Figure 1a and b displays that both the NC and B-NC samples have a spherical morphology with a diameter of ~12 μm. XRD patterns (Figure S1, Supporting Information) illustrate that both the fresh samples have the rhombohedral α-NaFeO₂ structure with the R $\bar{3}$ m space group without any impurity phase observed. The splitting of (006)/(102) and (108)/(110) peaks suggests that both samples have a well-formed layered structure. Figure S2 shows the Rietveld refinement results for the fresh NC and B-NC samples, and the excellent fitting of the simulated and experimental XRD patterns indicates that the refinement results are convincing. Table S1 suggests a slight expansion of the lattice parameters and unit cell volume of B-NC compared to NC, which agrees well with the results in the previous studies of boron-based polyanion-tuned high-Ni/Li-rich layered cathodes and implies a doping of some atomic boron into the host lattice structure.^{4,30,32} The unit cell expansion is conjectured to be caused by the occupancy of the small B³⁺ ions ($r_{B^{3+}} = 0.27 \text{ \AA}$) in the tetrahedral interstitial sites of oxygen packing of both Li and transition-metal (TM) layers.^{4,30}

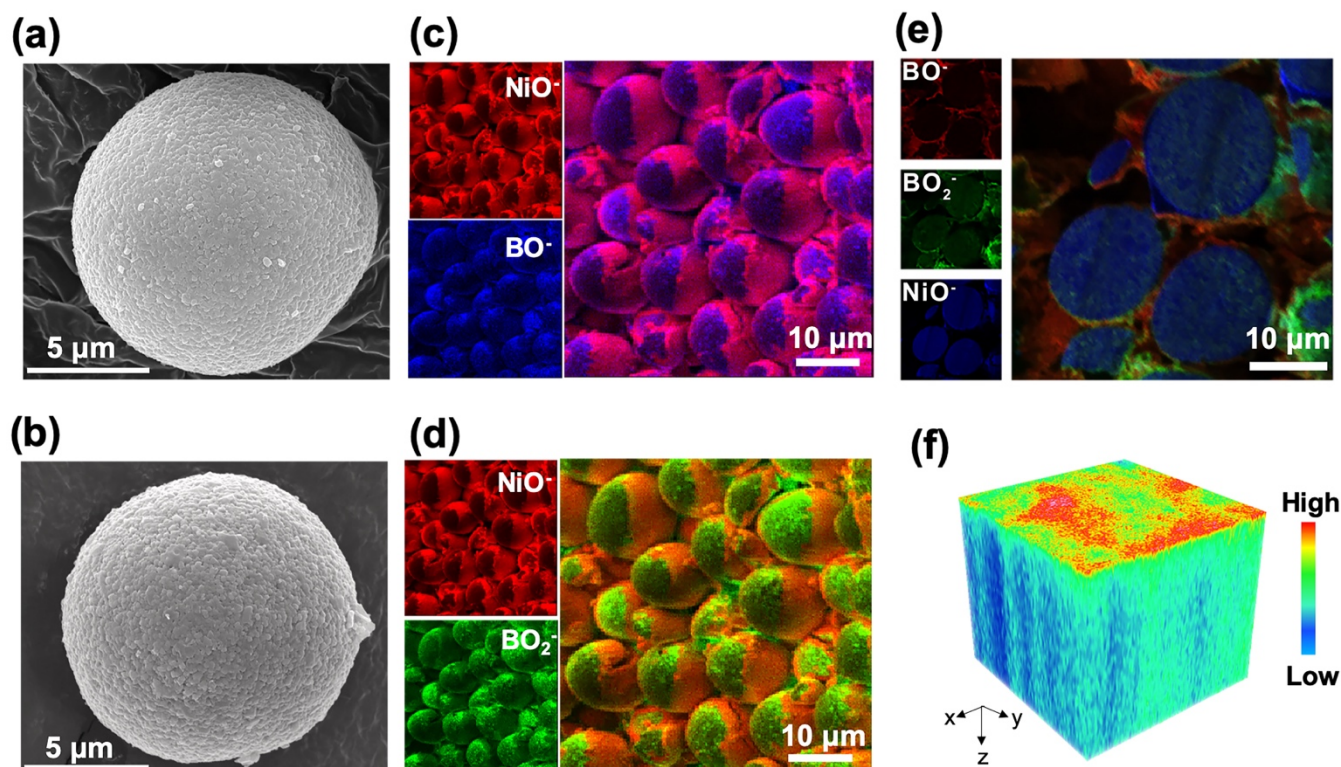


Figure 1. SEM images of (a) fresh NC and (b) fresh B-NC. TOF-SIMS surface mapping of (c) BO⁻ and NiO⁻ species and (d) BO₂⁻ and NiO⁻ species in fresh B-NC particles. (e) TOF-SIMS cross-section mapping of BO⁻, BO₂⁻, and NiO⁻ species in fresh B-NC. (f) TOF-SIMS three-dimension depth profiling of BO_x⁻ ($x = 1$ or 2) species in the fresh B-NC particle.

To identify, locate, and relatively quantify the species the inserted boron generates in the host material, we performed TOF-SIMS depth profiling and high resolution imaging on the fresh B-NC sample. Figure 1c - e and Figure S3 clearly indicate the presence of certain amount of boron oxides (represented by BO⁻ and BO₂⁻, hereafter termed as BO_x⁻) and trace amount of LiBO₂⁻ (probably resulting from the reaction between B₂O₃ and LiOH during calcination) in the fresh B-NC. The cross-section mapping and depth profiling (Figure 1e - f and Figure S3) show that

most of the detected BO_x⁻ species are present at the very surface of the particle with a low-concentration distribution in the interior, while the produced LiBO₂⁻ is predominately present on the surface. Note that, despite their presence, BO_x⁻ and LiBO₂⁻ do not influence the structure of the bulk material probably due to their limited concentrations. To conclude, TOF-SIMS and XRD reveal residual boron oxides and lithium borates in the boron-based polyanion-tuned layered oxides, in addition to the atomic boron doped into the host lattice structure.

3.2 Comparison of Electrochemical Performance

Electrochemical performance of both the samples were evaluated in both half cells (vs Li/Li⁺) and pouch-type full cells (vs graphite), and the results are presented in Figure 2. Figure 2a shows the capacity-cycle curves of both the samples in half cells. After an activation of ~ 6 - 10 cycles, the capacity of both the samples reach their maximum values of, respectively, ~ 226 mA h g⁻¹ for NC and ~ 223 mA h g⁻¹ for B-NC. The marginal capacity difference suggests that the added B₂O₃ causes negligible capacity loss, consistent with other studies.^{4,30} But the capacity retention of B-NC is 85% after 200 cycles, remarkably exceeding the 68% of NC. Figure 2b and c display the evolution of the charge-discharge curves for both the samples over the course of 200 cycles. Both samples show similar charge-discharge curves at the beginning of cycling, in which the plateaus corresponding to the phase transitions H1 → H2 → H3 are discernible (first cycle charge-discharge curves shown in Figure S4a).³³ But the plateaus of NC decay more rapidly than those of B-NC over cycling. The average difference between the charge and discharge voltages (Figure S4b and c) for NC is 0.34

V, which is higher than the 0.21 V found for B-NC after 200 cycles. The rapid voltage fade for NC results from a rapid increase in impedance upon cycling, which results from the transition metal dissolution and the thickening of CEI and SEI layers on cathode and anode, respectively (this will be discussed in the following section). All these results imply that B-NC has a better energy retention. Figure 2d illustrates the rate performance of both the samples at different current densities. The similar discharge capacity of both the samples with an increase in current density suggests that the incorporation of boron oxide does not affect the rate performance. Both samples deliver a high capacity of ~ 170 mA h g⁻¹ at 10C rate. To evaluate the practical viability of B-NC, the long-term cyclability of both samples were examined in pouch-type full cells paired with a graphite anode, and the results are displayed in Figure 2e and f. The capacity and average discharge voltage retention of B-NC is, respectively, 80% and 96% after 400 cycles at C/2 rate, substantially superior to the 61% and 93% of NC (more pouch cell data are shown in Figure S4d). The improved electrochemical performance of B-NC in both coin cell and pouch full cell therefore calls for an understanding of the mechanism.

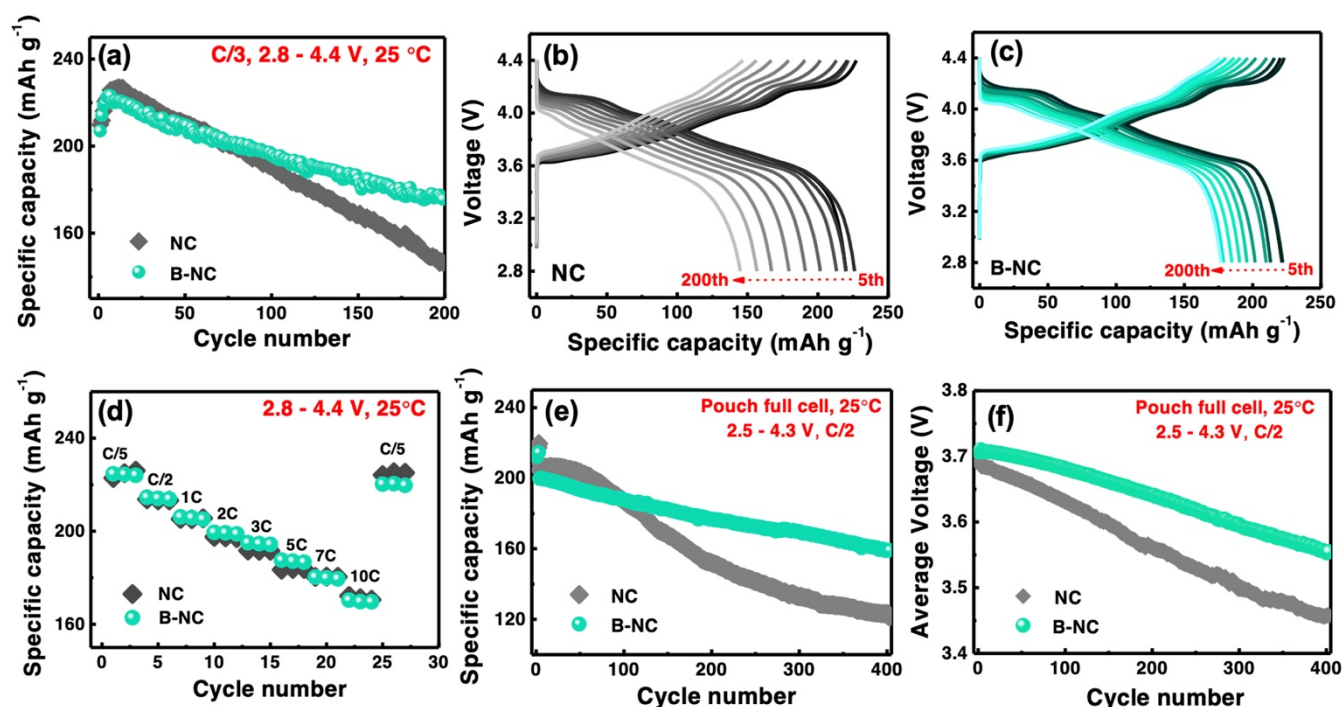


Figure 2. Electrochemical performance comparison of NC and B-NC. (a) Cycling performances of both samples in half cell (vs Li/Li⁺). Charge-discharge curves evolution of (b) NC and (c) B-NC over the course of 200 cycles. (d) Rate performance of NC and B-NC in half cells (vs Li/Li⁺). (e) Cycling performance comparison of both samples in pouch full cell with graphite anode cycled between 2.5 and 4.3 V at room temperature. (f) Average discharge voltage of pouch full cells with NC and B-NC over the course of 400 cycles.

3.3 Evaluation of Structure and Phase Transition

To find out all of the possible contributions to the enhanced electrochemical performance of B-NC, the structural evolution of both cathodes upon charging was studied via *in-situ* XRD. Figure 3a presents the shifting of the (003) Bragg reflection of both the cathodes upon charging (the results for other peaks are shown in Figure S5). Both materials exhibit a similar phase evolution upon charging (H1 → H2 → H3). Specifically, the (003)_{H1} peak smoothly shifts to a lower 2θ angle upon charging and finally evolves into (003)_{H2} at ~ 4.0 V. But further charging results in a sharp shift of the (003)_{H2} peak backward to a higher

2θ, accompanied by the formation of the H3 phase as well as the phase separation of H2 and H3 phases from ~ 4.2 V to the end of charge. Besides, the lattice parameters of both samples exhibit similar contraction by the end of charge, 2.0 % and 5.6 %, respectively, for the *a* and *c* axes (Figure S6). These results suggest that boron-based polyanion from B₂O₃, despite presenting in the host structure and on the surface of material, does not suppress the phase transition and lattice anisotropic changes.⁴ Severe anisotropic lattice changes are thought to be one of the major reasons for the rapid capacity fading of high-Ni cathodes, and alleviating the lattice contraction has been proven effective in improving the capacity retention.^{8,34-36} The current *in-situ*

XRD results, however, show that the incorporation of boron-based polyanion does not mitigate the phase transitions upon cycling, eliminating the direct correlation between the improved cyclability of B-NC and the lattice structure stability. Previous study proposed that the improved cyclability of B-doped $\text{LiNi}_{0.9}\text{Co}_{0.05}\text{Mn}_{0.05}\text{O}_2$ could be due to alterations in the primary particle array resulting in an optimized secondary-particle microstructure.⁴ In view of this, cross-sectional SEM investigations were performed on both the cathodes before and after 400 cycles in pouch full cells. Figure 3b and Figure S7 show the cross-section SEM images of NC and B-NC before and after cycling, respectively. Careful examinations of these images indicate that the overall primary particle array in both cathodes is radial and that there is no significant difference in the secondary particle microstructure of both the samples before and after cycling. These results signify that the incorporation of boron-based polyanion has little effect on the primary particle array and the mechanical structure of B-NC.

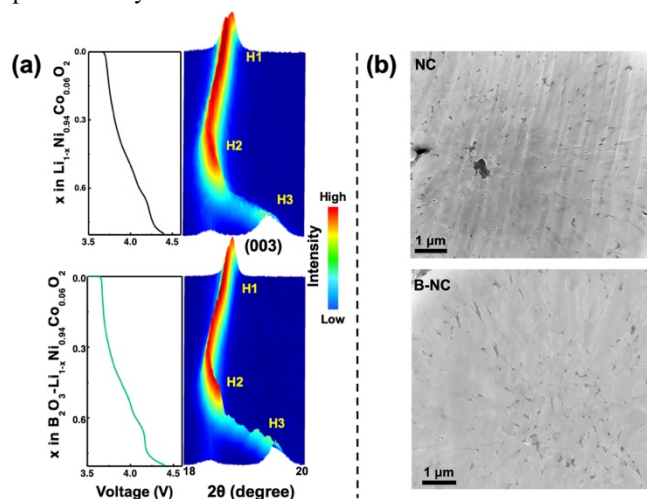


Figure 3. (a) *In-situ* XRD contour plot of (003) peak of NC and B-NC charged at C/10 rate and the corresponding voltage-state of delithiation curves. (b) Cross-section SEM images of fresh NC and B-NC samples.

3.4 Investigation on Cathode/Anode Electrolyte Interphase

Besides the consideration of the structural factors, the presence of $\text{BO}_x^-/\text{LiBO}_2^-$ on the surface of B-NC prompts an all-inclusive investigation on the chemistries of the cathode-electrolyte interphase (CEI) and the graphite anode solid-electrolyte interphase (SEI). For high-Ni cathodes, surface residual lithium could influence the surface chemistries of the electrodes during electrochemical cycling. It is necessary to compare the surface residual lithium of both the fresh samples first. The chemical titration results in Figure S8 show that the residual lithium content is very close in the fresh NC and B-NC, indicating that the possible surface chemistries related to residual lithium for NC and B-NC should be comparable during cycling.

Ultrahigh chemical-selectivity TOF-SIMS technique can provide a high-sensitivity detection of different elements, particularly the light ones, such as Li and B, enabling a detailed investigation into the interphase of both the cycled cathodes in pouch full cells. Figure 4a plots the depth profiles of the representative CEI species, including the non-metal interphase species (C_2HO^- , C_2F^- , PO_3^- , and POF_2^- , characterizing the decomposition of the electrolyte solvent and LiPF_6) and the dissolved metal-based species ($^7\text{LiF}_2^-$, NiF_3^- , and CoF_3^-).⁹ As plotted, the

CEIs of both samples consist of two layers, an outer layer with diverse non-metal species and an inner layer with different metal-based species; the comparison of the distribution profiles of different species shows that the CEI of B-NC is much thinner than that of NC after 400 cycles.

A detailed comparison of the representative non-metal and metal-based species in Figure 4b and Figure S9 shows significant differences between the CEIs of both cathodes. Specifically, diverse kinds of non-metal species (C_2F^- , PO_3^- , C_2P^- , CHO^- , POF_2^- , etc.) dominate at the surface of both CEIs. But the non-metal species of B-NC, particularly the phosphorus-bearing species (originating from the complex reactions between LiPF_6 , electrolyte solvents, $\text{BO}_x^-/\text{LiBO}_2^-$, H_2O , and cathode materials), have significantly stronger intensity than those of NC and concentrate in a narrower range, implying a more compact and passivated non-metal phosphorus-rich outer layer formed in the CEI of B-NC. Closely following the outer, non-metal layer is the inner layer consisting of diverse dissolved metal-related species ($^7\text{LiF}_2^-$, NiF_3^- , CoF_3^- , etc.), as shown in Figure 4 and Figure S9c. Contrary to the non-metal species, the dissolved metal-bearing species in NC have a much stronger intensity and wider distribution range than B-NC. These dissolved metal species are highly-resistive to electronic and Li-ion migration, and are considered as a major reason for the voltage and capacity decay as well as the rapid growth of cell impedance.^{6,9,37} Besides depth profiling, we used TOF-SIMS chemical mapping of the representative species, NiO^+ , $^7\text{LiF}_2^-$ and C_2F^- (Figure 5a) to visualize the lateral distribution of the CEI on both cycled samples. As shown, the majority of both samples are covered by the CEI species of $^7\text{LiF}_2^-/\text{C}_2\text{F}^-$ after a short sputtering (100 s). With the sputtering extended to 1 h, the bulk cathode material (represented by NiO^+) is disclosed.

Besides, it is necessary to figure out whether the $\text{BO}_x^-/\text{LiBO}_2^-$ species on the particle surface are involved in the CEI formation of B-NC upon cycling. To address this question, the combined TOF-SIMS positive (O_2^+ sputtering) and negative (Cs^+ sputtering) polarity detection modes were used to scrutinize the cycled B-NC cathode. Figure 5b presents the detected boron-related species as well as their three-dimensional depth profiling. As plotted, several kinds of boron-bearing species were detected, including BO_x^- , BOF^+ , and BF_y^+ ($y = 2$ or 3), all predominately concentrated at the very surface of the cycled B-NC (~ 5 nm). The presence of BO_x^- suggests residual boron oxides in B-NC after 400 cycles. The BF_y^+ species probably originate from the reactions of boron oxides with HF, which follows the reaction of LiPF_6 with H_2O during cycling. The presence of BOF^+ species implies that some complicated reactions between BF_3^+ and H_2O (from the cathode or electrolyte) may occur during cell cycling.³⁸ Previous investigations unambiguously revealed that the B-O/B-F-containing species can enhance the robustness of the cathode CEI and thus be beneficial to the cycling performance.³⁹⁻⁴¹ The currently detected boron-bearing species on the surface of the cycled B-NC indicate that the CEI of B-NC is boron-rich and strongly related to the improved cyclability of B-NC.

The graphite anodes cycled with both cathodes for 400 cycles were also examined by TOF-SIMS. Figure S10 presents the depth profiles of the representative SEI species in both graphite anodes, and the comparison indicates that the SEI of the graphite paired with NC is thicker than that with B-NC. Depth profiling of different species in Figure S10 and Figure S11 unveil more details about the SEIs. First, the SEI of the graphite paired with NC has a much higher concentration of the deposited TM-

and Li-related species than that with B-NC (represented by $^7\text{Li}^+/\text{LiF}^-$ and NiF_3^- in Figure S10 as well as $^6\text{Li}^+/\text{LiO}_2^-$ and $^{58}\text{Ni}^+/\text{Co}^+/\text{CoF}_3^-$ in Figure S11a). Considering that the cathode is the only source of the dissolved transition metal ions, the decreased concentration of TMs in graphite paired with B-NC further strengthens the assertion that the well-formed CEI in B-NC inhibits the dissolution and crossover of transition metal ions.

Second, the SEI of the graphite paired with NC also shows increased concentration of the solvent and LiPF_6 decomposition products (characterized by CHO_2^- , $\text{C}_2\text{H}_3\text{O}^-$, C_2OF^- , PO_2^- , C_2P^- , POF_2^- etc., in Figure S10 and S11b-c). Note that no boron-bearing species were detected by TOF-SIMS from the cycled graphite that was coupled with B-NC in this study.

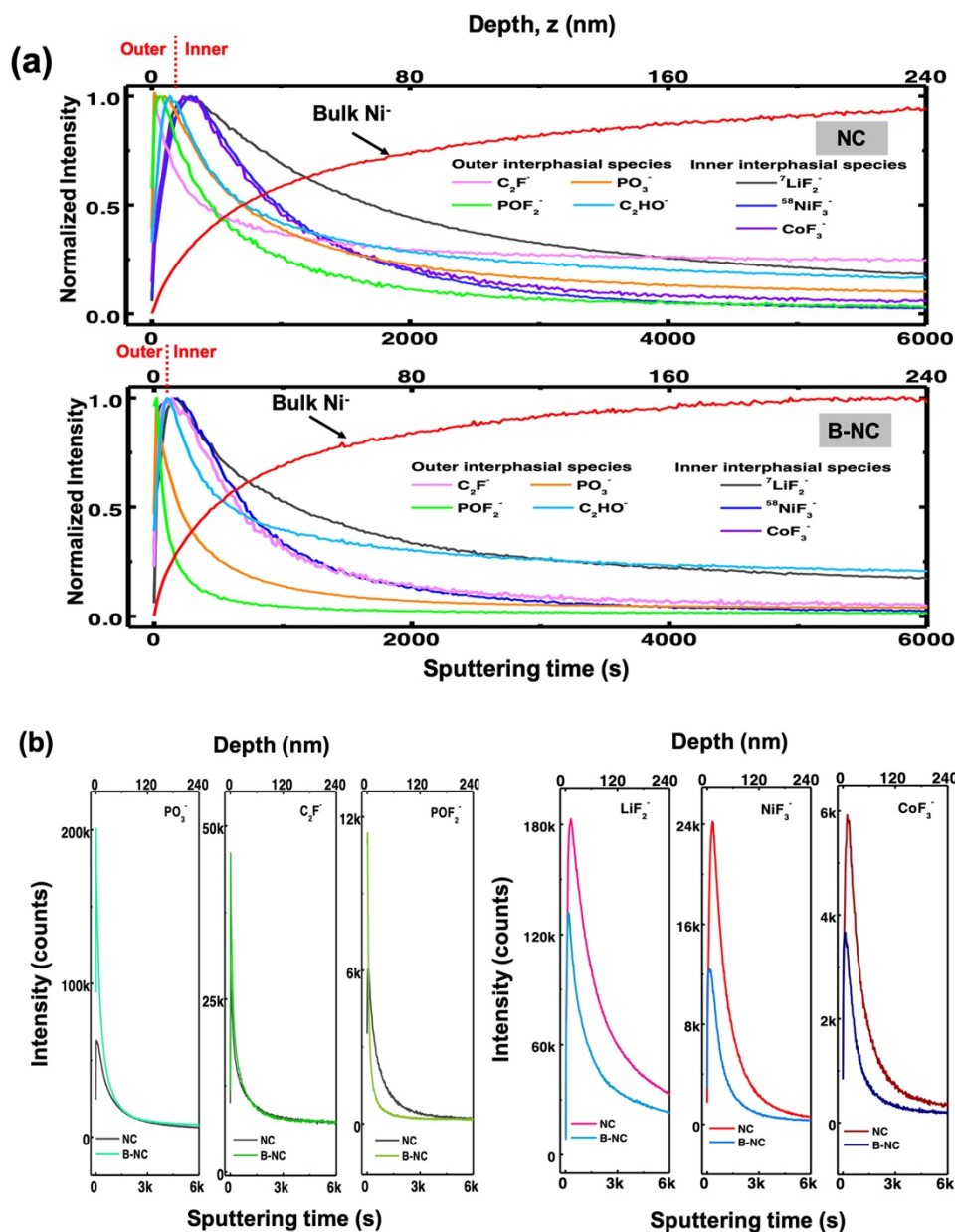


Figure 4. (a) TOF-SIMS depth profiling of the characteristic cathode/electrolyte interphase (CEI) species of the NC and B-NC sample cycled in pouch full cells for 400 times. (b) TOF-SIMS intensity and distribution comparison of the representative inorganic non-metal and metal-based species in the CEI of the NC and B-NC sample cycled in pouch full cells for 400 time.

Previous studies showed that the deposition of transition metals onto the graphite anode could destroy the integrity and stability of the formed SEI, intensify the accumulation of lithium dendrites, and induce a continual growth of fresh SEI by consuming electrolyte and trapping lithium inventory.^{11,24,42} Therefore, the increased TM deposition on the graphite with NC leads to more electrolyte decomposition products and irreversible lithium (dead lithium, as indicated by the Li^+ depth profiling

shown in Figures S10 and S11) deposited on the graphite surface. Those deposited Li could deteriorate the capacity and energy retention, as evidenced by the cycling performance of NC in Figure 2, and induce potential internal short circuit and safety issues.^{16,42}

XPS investigations provide additional information about the surface chemistry of both cycled cathodes and anodes. Figure S12 compares the elemental concentrations of both cycled

cathodes. The higher concentration of carbon in B-NC may suggest more carbon-containing species on the surface of B-NC cathode, and the increased phosphorus concentration in B-NC indicates a phosphorus-rich surface and agrees well with the TOF-SIMS results. The increased C-C bonding (284.8 eV, characteristic of carbon black) in the C 1s spectrum of B-NC (Figure 6a) implies a thinner CEI since a thinner CEI enables more C-C signal exposed from the conductive carbon black and thus an elevated total carbon content.⁴⁰ Also, the increased C species at ~ 286.5 eV (originating from C-O-type species/PVDF binder),

which aligns with the increased C-O bonding (~ 534 eV, also originating from C-O species) in the O 1s spectrum (Figure 6b), suggests that the CEI of B-NC probably consists of more C-O-containing decomposition products, such as Li-alkoxides and polymeric C-O compounds.⁴⁰ Besides, the B-O peak (~ 192.5 eV) representing the B_xO_y species is observed in the B 1s spectra of B-NC (Figure 6c), consistent with the TOF-SIMS results.^{39,43} Additional cathode XPS data are shown in Figure S13.

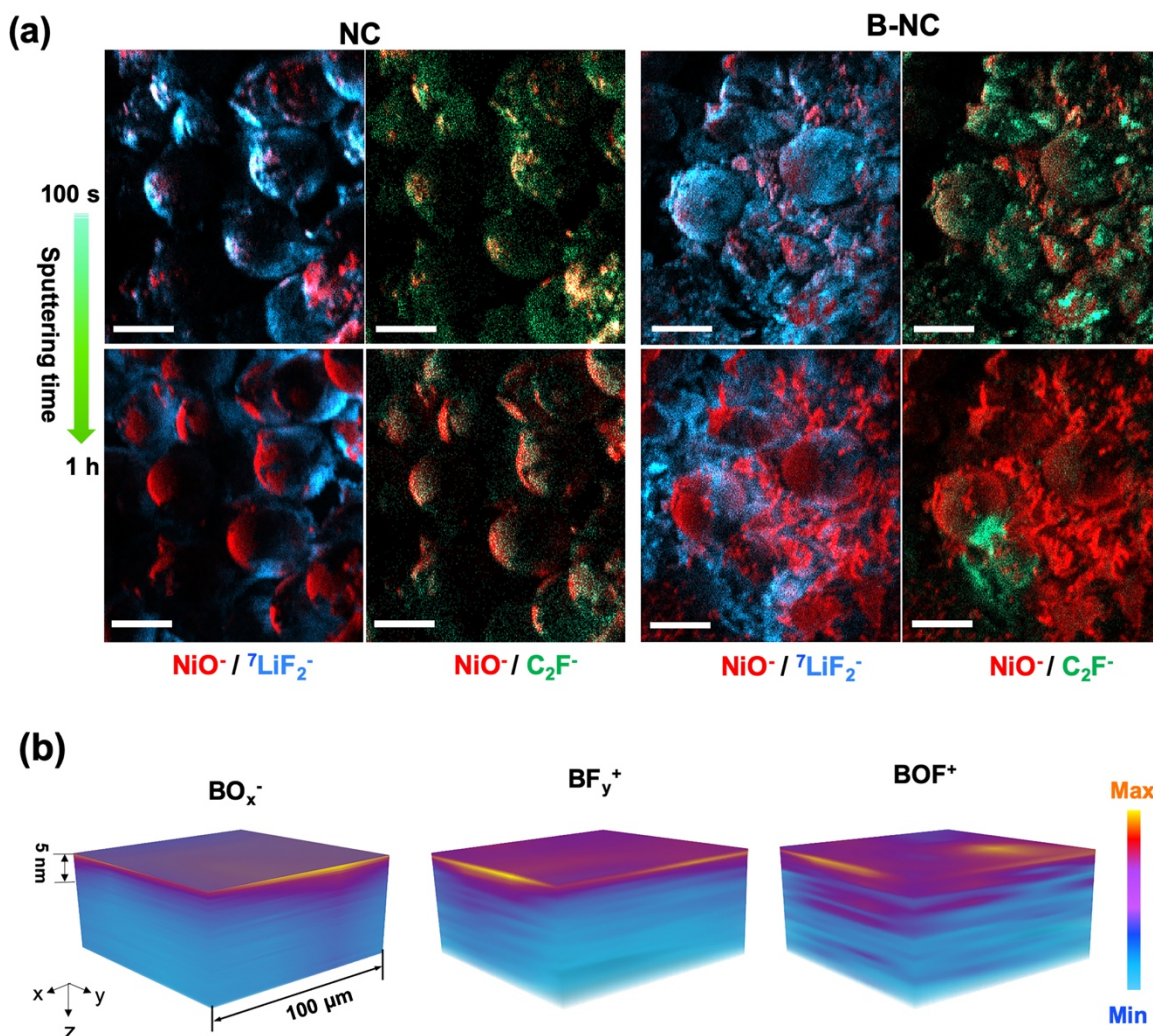


Figure 5. (a) TOF-SIMS mapping of the surface evolution of the 400-cycled NC and B-NC samples in pouch full cell with the sputtering time extending from 100 s to 1 h. (b) Three-dimensional depth profiling of the detected boron-related species in the B-NC sample after 400 cycles in pouch full cell. The scale bar in (a) is 10 μm .

XPS results of the graphite anodes paired with both the cathodes are exhibited in Figures S14 and S15. As illustrated in Figure S14, the graphite paired with B-NC displays an elevated concentration of C, but reduced concentrations of O, P, Li, and Ni than that with NC. The XPS C 1s spectrum of the graphite paired with B-NC (Figure S15) shows a higher concentration of C-C bonding (284.8 eV) than that paired with NC, suggesting a thinner SEI on the graphite paired with B-NC since the dominant source of the C-C bonding in the anode is the bulk graphite. Also, the concentration of the C=O bonding in the C 1s spectrum of the B-NC graphite is increased compared to that with NC, concordant with the elevated concentration of C=O

bonding in the O 1s spectrum of the graphite paired with B-NC. This infers that the SEI of B-NC graphite may contain more C=O bearing species. Besides, the elevated concentrations of Li, P, Ni, and LiF/TMF_x species in the graphite paired with NC (Figure S14 - S16) reflect an increased deposition of metal-based species and the decomposition of electrolyte, verifying the TOF-SIMS results.

Collectively, the structure and interphase analysis indicate that the formation of a compact and well-passivated boron/phosphorus-rich CEI in B-NC contributes predominately to the improved cyclability of B-NC. Based on the TOF-SIMS and

XPS results, the formation pathway of the CEI for NC and B-NC is proposed as follows: with both samples cycled and the CEI formed via the electrode/electrolyte interactions, LiPF_6 in the electrolyte reacts with trace amounts of H_2O in the electrolyte/cathode to produce corrosive HF. The produced HF in NC likely destroys the formed CEI layer repeatedly and corrodes the active material, resulting in a thickened, porous CEI and serious TM dissolution and crossover. However, for B-NC, the presence of $\text{BO}_x/\text{LiBO}_2$ species on the surface of B-NC function as a scavenger of HF and H_2O , consuming HF and H_2O and thus stabilizing the formed CEI and alleviating the corrosion of active material. Meanwhile, the produced boron-bearing reactants (BF_3^+ , BOF^-) in B-NC are actively *in-situ* involved in the generation of CEI upon cycling, contributing to a compact and well-passivated boron/phosphorus-rich CEI. Besides, the suggested presence of boron polyanion (atomic boron) in the bulk

lattice structure of B-NC may also help reduce the TM dissolution. Specifically, the enhanced oxygen stacking stability by the boron polyanion, as discussed in section 3.1, may help alleviate the surface structure reconstruction from layered to rocksalt structure in the B-NC particle, thereby reducing the TM dissolution.¹¹ But it should be noted that the current characterization results indicate that the *in-situ* formed CEI is more important in improving the cyclability of B-NC. Overall, the well-passivated CEI, along with the proposed inhibition of structure reconstruction, alleviate the severe corrosion of active material upon long-term cycling, reduces the TM dissolution, crossover, and deposition onto the graphite anode, and suppresses the loss of active lithium, which together account for the significantly enhanced cyclability of B-NC.

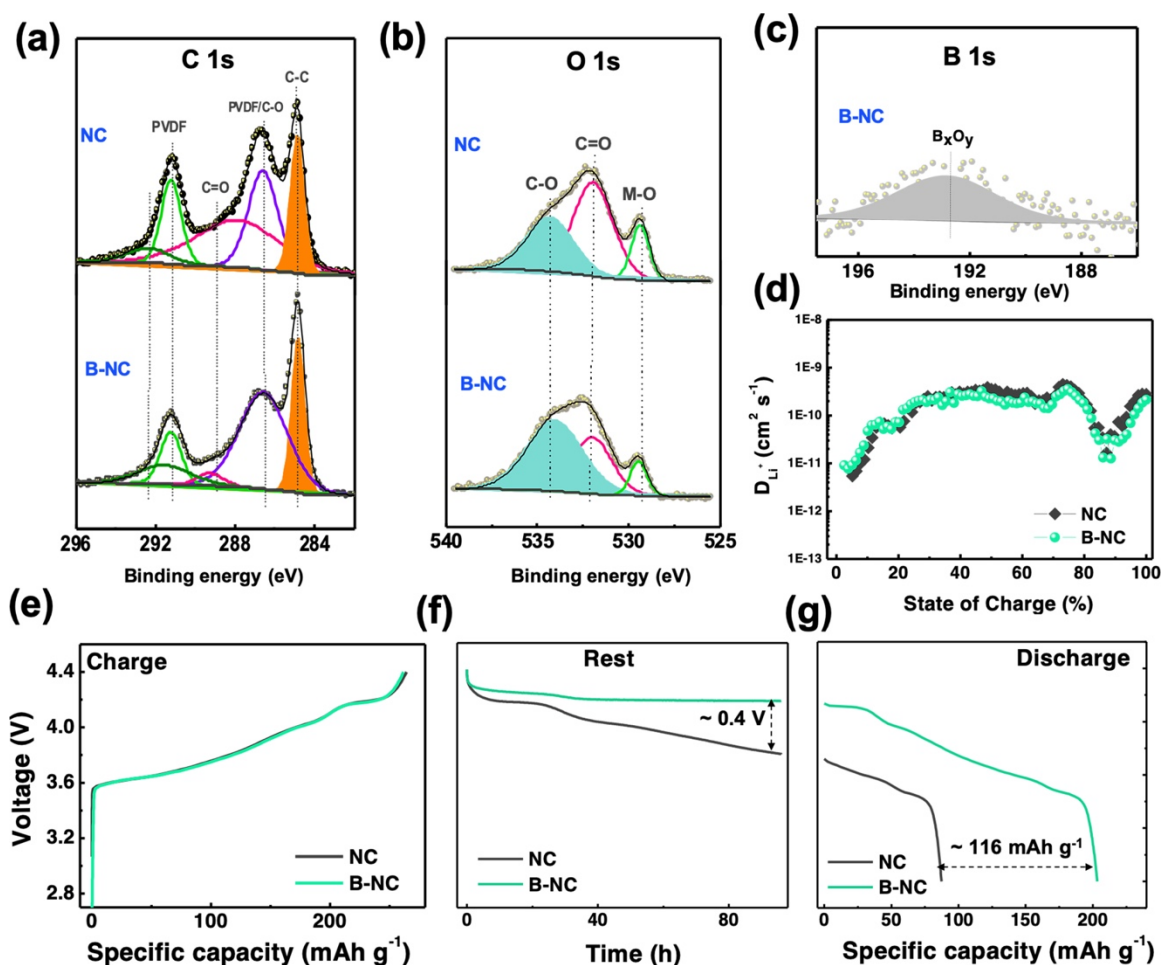


Figure 6. XPS data of (a) C 1s and (b) O 1s of the NC and B-NC samples cycled in pouch full cells for 400 times. (c) B 1s XPS data of the cycled B-NC sample in pouch full cells for 400 times. (d) Li^+ diffusion coefficient (D_{Li^+}) of the NC and B-NC samples as a function of the state of charge. Self-discharge test of NC and B-NC vs Li/Li^+ at 55 °C: (e) the charge curve, (f) changes of the open-circuit voltage during the dwelling of 96 h, and (g) the discharge curve after dwelling.

In addition to the survey on interphase chemistry, the ionic conductivity and the surface stability of both the samples under high-voltage and high-temperature conditions were also evaluated by galvanostatic intermittent titration technique (GITT) and self-discharge test. GITT results in Figure 6d reveal that both cathodes deliver a similar change in conductivity magnitude, indicating that the incorporation of boron-based polyanion does not adversely affect the ionic migration upon cycling,

agreeing with the rate performance in Figure 2d as well (GITT charge curves of both samples are plotted in Figure S17). For the self-discharge test at 55 °C, coin cells with both cathodes were first charged to 4.4 V (vs Li/Li^+) at C/3 rate, followed by a rest of 96 h at 55 °C, and then discharged to 2.8 V at C/3 rate. By the charging endpoint of 4.4 V, both materials deliver a similar charge capacity (Figure 6e). However, the open circuit voltages (OCV) of both charged samples show a significant

difference over the dwelling of 96 h as illustrated in Figure 6f. The cell with NC experiences a continuous OCV drop and ends up at ~ 3.76 V, suggesting a rapid self-discharge during rest. In contrast, the cell with B-NC almost maintains a constant OCV of ~ 4.16 V after a slight initial drop. After resting, the cell with NC only offers a discharge capacity of 87 mA h g^{-1} , substantially lower than the 203 mA h g^{-1} of B-NC. These results imply that the CEI of the B-NC sample is more stable and resistant to the unfavorable electrode/electrolyte reactions under high voltage and high temperature conditions.

3.5 Improved Air-exposure and Thermal Stability by Boron-based Polyanion

Air sensitivity is another concern in the practical application of high-Ni cathodes. To evaluate the air sensitivity of both the samples, the fresh and 30-day air-stored NC and B-NC were examined by the residual lithium titration and electrochemical tests. XRD patterns in Figure S18 illustrate that the stored NC and B-NC retain the $\alpha\text{-NaFeO}_2$ structure with a R3m space group and that no peaks representing LiOH or Li_2CO_3 are observed, inferring that the bulk structure is maintained after 30-day air storage.

Figure 7a and b present the changes of the residual lithium content in NC and B-NC samples before and after storage. Despite a similar amount of residual lithium for both the fresh cathodes, the total residual lithium content in NC increases 13% after the 30-day air storage, in sharp contrast to the 4% increase in B-NC. SEM images in Figure S19 clearly show a discrepancy in the surface morphology of both the stored cathodes. In addition, the content of Li_2CO_3 in NC increases dramatically after the storage, accompanied by a decrease in LiOH. In contrast, the content of $\text{Li}_2\text{CO}_3/\text{LiOH}$ in B-NC shows marginal changes. This data indicates that B-NC is more chemically stable to ambient air, which could be explained by the following reasons. First, the presence of the BO_x^- and LiBO_2^- on the very surface of B-NC particles (as illustrated by TOF-SIMS) may hinder the direct contact of active material with air, thus reducing the occurrence of side reactions with the environment. Second, previous DFT calculation and X-ray absorption spectroscopy studies have revealed that the incorporation of boron-based polyanion into layered oxide cathodes could stabilize the oxygen packing of host materials and introduce strong B-O bonds (bond enthalpy: 809 kJ mol^{-1}).^{4,28,32} These could prevent the loss of lattice oxygen from the bulk structure, a major reason for the degradation of high-Ni cathode upon air exposure,¹ and suppress the undesired reactions between high-Ni cathodes and $\text{H}_2\text{O}/\text{CO}_2$ in air and thereby improving the storage stability of B-NC.

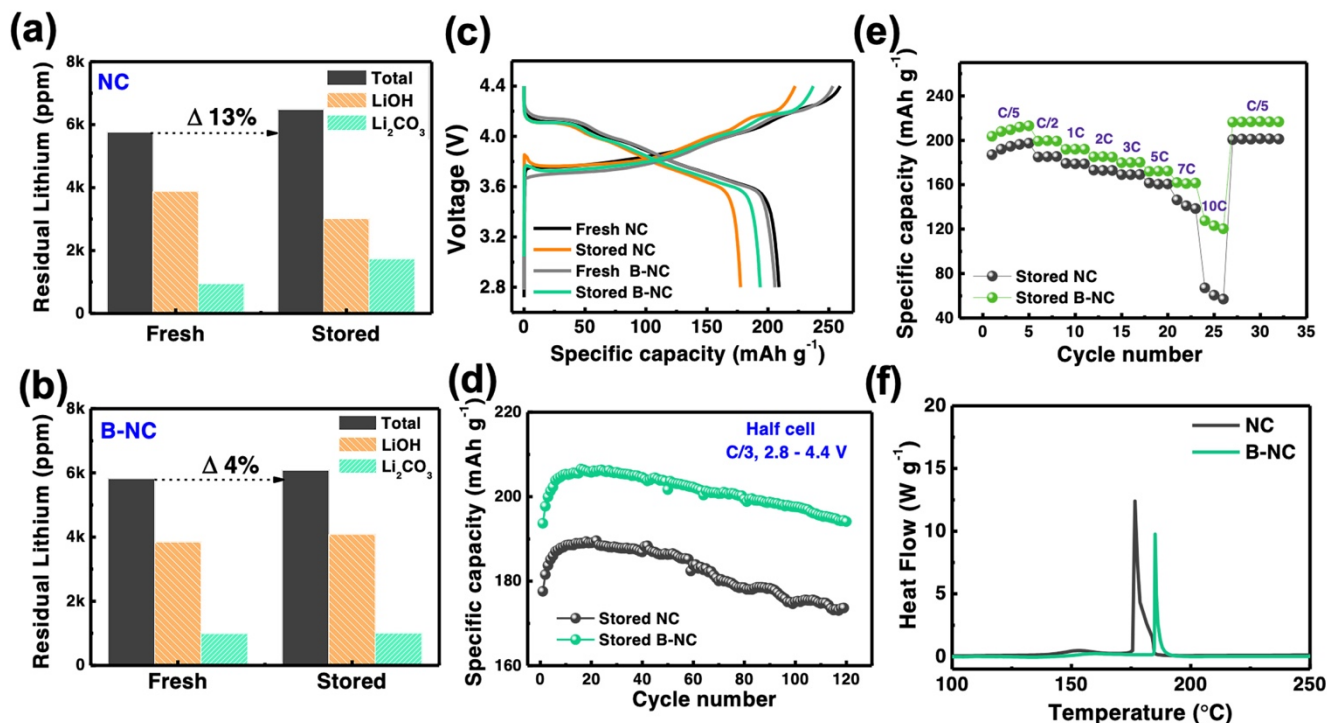


Figure 7. Residual lithium content changes of the fresh and 30-day air-stored (a) NC and (b) B-NC. (c) First charge-discharge curves comparison of the fresh and 30-day stored NC and B-NC. (d) Cycling performance comparison of the 30-day stored NC and B-NC in half coin cells (vs Li/Li^+). (e) Rate performance of the 30-day stored NC and B-NC over the course of C/5 to 10C rate. (f) DSC results of the fresh NC and B-NC.

The electrochemical performance of the stored NC and B-NC is also compared in Figure 7. The first cycle charge-discharge curves in Figure 7c illustrate that the stored NC shows significant charging polarization and significantly reduced discharge capacity (177 mA h g^{-1} , only 84% of the initial capacity of the fresh sample). In contrast, the stored B-NC delivers an initial capacity of 194 mA h g^{-1} (94% of the capacity of the fresh B-NC) with modest charging polarization. The cycling performance comparison in Figure 7d further shows the superior

electrochemical performance of the stored B-NC. The maximum capacity and capacity retention of the stored B-NC is, respectively, 206 mA h g^{-1} and 94% after 120 cycles while it is only 190 mA h g^{-1} and 91% for the stored NC. The rate performance of the stored B-NC also outperforms the stored NC (Figure 7e) from C/5 to 10C rate. Particularly, the stored B-NC delivers a discharge capacity of $\sim 125 \text{ mA h g}^{-1}$ at 10C rate, almost twice that of the stored NC at 10C rate ($\sim 65 \text{ mA h g}^{-1}$). The well-maintained electrochemical performance of the stored B-

NC is primarily ascribed to its reduced air sensitivity and the suppressed residual lithium increase since the suppressed residual lithium increase can reduce the surface delithiation layer and the loss of the active Li^+ content from the bulk material, lower the internal resistance, and inhibit the unfavorable interactions between electrolyte and residual lithium species.^{1,14,32,44}

Differential scanning calorimeter (DSC) test was performed to evaluate the thermal performance of both the samples, which is an indicator of the battery safety and is of paramount importance to the practical viability of high-Ni cathodes. Figure 7f shows that the exothermic peak temperature and the released heat are, respectively, 177 °C and $\sim 1700 \text{ J g}^{-1}$ for NC and 185 °C and $\sim 1250 \text{ J g}^{-1}$ for B-NC. Following proposes an understanding of the improved thermal stability of B-NC. Thermal stability studies by Ceder *et al* suggested that the increased thermal stability of layered high-Ni cathodes reflects the strengthened resistance of the charged cathode to oxygen release from the bulk structure at high temperature.^{17,45,46} For the highly delithiated layered cathode $\text{Li}_{1-x}\text{Ni}_{0.94}\text{Co}_{0.06}\text{O}_2$ ($x > 0.5$), the release of O_2 at elevated temperature is closely associated with the conversion of the layered phase into the spinel/rocksalt phases (LiM_2O_4 , M_3O_4 , MO , $\text{M} = \text{Ni/Co}$), which depends on the migration of Ni/Co from TM layer to the octahedral sites of Li layer by passing through the neighboring tetrahedral sites.¹⁷ In the charged B-NC, the presence of the B^{3+} ions in the tetrahedral interstices as proposed by Pan and our XRD results may impede the migration of Ni ions.³⁰ Also the boron doped into the lattice structure was suggested to enhance the oxygen packing stability and provide stronger B-O bonds, thereby suppressing the release of oxygen at elevated temperatures.^{4,28} The suppressed migration of Ni/Co and oxygen release contribute together to the improved exothermic temperature of B-NC. In addition, the charged B-NC likely decomposes to boron oxides upon heating, which reduces the available oxygen to react with the electrolyte and accounts for the decreased heat release.

4. Conclusion

This work presents a comparative study of $\text{LiNi}_{0.94}\text{Co}_{0.06}\text{O}_2$ (NC) and $\text{B}_2\text{O}_3\text{-LiNi}_{0.94}\text{Co}_{0.06}\text{O}_2$ (B-NC), providing fundamental insights into boron-based polyanion-tuned layered oxides. By employing advanced three-dimensional chemical analysis, we identify, locate, and relatively quantify various species of interest that form during preparation and cycling of these materials. The spatial distribution of the boron source is established. We find $\text{BO}_x/\text{LiBO}_2^-$ species on the surface and inside of the B-NC particles, in addition to the doping of atomic boron into the bulk lattice structure. Structure and interphase characterizations clearly demonstrate that the boron-bearing species on the particle surface contributes to a compact and well-passivated boron/phosphorus-rich CEI in B-NC, which is predominately responsible for the improved capacity retention of 80% for B-NC after 400 cycles in full cell with graphite anode (61% for NC). Meanwhile, the presence of the doped boron polyanion (atomic boron) in the lattice structure of B-NC accounts for the improved air-exposure stability and thermal stability by enhancing the oxygen structure stability, with the B-NC stored for 30 days in air maintaining a high capacity of 125 mA h g^{-1} at 10C rate (only 65 mA h g^{-1} for stored NC). Overall, the present study complements and extends the understanding of boron-based polyanion-tuned layered oxides by a multi-perspective investigation and offers significant practical promise for these systems.

ASSOCIATED CONTENT

Supporting Information. The supporting information includes XRD patterns and Rietveld refinement results, electrochemical data, TOF-SIMS data, XPS results, and SEM images *etc.* This material is available free of charge on the ACS Publication website.

AUTHOR INFORMATION

Corresponding Author

* (A. M.) E-mail: manth@austin.utexas.edu

ACKNOWLEDGMENT

This work was supported by the Assistant Secretary for Energy Efficiency and Renewable Energy, Office of Vehicle Technologies of the U.S. Department of Energy through the Advanced Battery Materials Research (BMR) Program (Battery500 Consortium) award number DE-EE0007762 and the Welch Foundation grant F-1254.

REFERENCES

- (1) You, Y.; Celio, H.; Li, J.; Dolocan, A.; Manthiram, A. Modified High-Nickel Cathodes with Stable Surface Chemistry Against Ambient Air for Lithium-Ion Batteries. *Angew. Chem.* **2018**, 130 (22), 6590–6595.
- (2) Sun, H.-H.; Manthiram, A. Impact of Microcrack Generation and Surface Degradation on a Nickel-Rich Layered $\text{Li}[\text{Ni}_{0.9}\text{Co}_{0.05}\text{Mn}_{0.05}]\text{O}_2$ Cathode for Lithium-Ion Batteries. *Chem. Mater.* **2017**, 29 (19), 8486–8493.
- (3) Li, J.; Li, W.; You, Y.; Manthiram, A. Extending the Service Life of High-Ni Layered Oxides by Tuning the Electrode-Electrolyte Interphase. *Adv. Energy Mater.* **2018**, 2, 1801957.
- (4) Park, K.-J.; Jung, H.-G.; Kuo, L.-Y.; Kaghazchi, P.; Yoon, C. S.; Sun, Y.-K. Improved Cycling Stability of $\text{Li}[\text{Ni}_{0.9}\text{Co}_{0.05}\text{Mn}_{0.05}]\text{O}_2$ Through Microstructure Modification by Boron Doping for Li-Ion Batteries. *Adv. Energy Mater.* **2018**, 4, 1801202.
- (5) Piao, J.-Y.; Sun, Y.-G.; Duan, S.-Y.; Cao, A.-M.; Wang, X.-L.; Xiao, R.-J.; Yu, X.-Q.; Gong, Y.; Gu, L.; Li, Y.; Liu, Z.-J.; Peng, Z.-Q.; Qiao, R.-M.; Yang, W.-L.; Yang, X.-Q.; Goodenough, J. B.; Wan, L.-J. Stabilizing Cathode Materials of Lithium-Ion Batteries by Controlling Interstitial Sites on the Surface. *CHEMPR* **2018**, 4 (7), 1–12.
- (6) Aurbach, D.; Markovsky, B.; Salitra, G.; Markevich, E.; Talyossef, Y.; Koltypin, M.; Nazar, L.; Ellis, B.; Kovacheva, D. Review on Electrode–Electrolyte Solution Interactions, Related to Cathode Materials for Li-Ion Batteries. *J. Power Sources* **2007**, 165 (2), 491–499.
- (7) Zhang, X.-D.; Shi, J.-L.; Liang, J.-Y.; Yin, Y.-X.; Zhang, J.-N.; Yu, X.-Q.; Guo, Y.-G. Suppressing Surface Lattice Oxygen Release of Li-Rich Cathode Materials via Heterostructured Spinel $\text{Li}_4\text{Mn}_5\text{O}_{12}$ Coating. *Adv. Mater.* **2018**, 30 (29), 1801751.
- (8) Xie, Q.; Li, W.; Manthiram, A. A Mg-Doped High-Nickel Layered Oxide Cathode Enabling Safer, High-Energy-Density Li-Ion Batteries. *Chem. Mater.* **2019**, 31 (3), 938–946.
- (9) Li, W.; Dolocan, A.; Oh, P.; Celio, H.; Park, S.; Cho, J.; Manthiram, A. Dynamic Behaviour of Interphases and Its Implication on High-Energy-Density Cathode Materials in Lithium-Ion Batteries. *Nature Communications* **2017**, 8, 14589.
- (10) Li, J.; Liu, H.; Xia, J.; Cameron, A. R.; Nie, M.; Botton, G. A.; Dahn, J. R. The Impact of Electrolyte Additives and Upper Cut-Off Voltage on the Formation of a Rocksalt Surface Layer in $\text{LiNi}_{0.8}\text{Mn}_{0.1}\text{Co}_{0.1}\text{O}_2$ Electrodes. *J. Electrochem. Soc.* **2017**, 164 (4), A655–A665.
- (11) Lin, F.; Markus, I. M.; Nordlund, D.; Weng, T.-C.; Asta, M. D.; Xin, H. L.; Doeff, M. M. Surface Reconstruction and Chemical Evolution of Stoichiometric Layered Cathode Materials for Lithium-Ion Batteries. *Nature Communications* **2017**, 8, 1–9.

- (12) Liu, H. S.; Zhang, Z. R.; Gong, Z. L.; Yang, Y. Origin of Deterioration for LiNiO₂ Cathode Material During Storage in Air. *Electrochem. Solid-State Lett.* **2004**, 7 (7), A190–A193.
- (13) Liu, H.; Yang, Y.; Zhang, J. Investigation and Improvement on the Storage Property of LiNi_{0.8}Co_{0.2}O₂ as a Cathode Material for Lithium-Ion Batteries. *J. Power Sources* **2006**, 162 (1), 644–650.
- (14) Chen, Z.; Wang, J.; Huang, J.; Fu, T.; Sun, G.; Lai, S.; Zhou, R.; Li, K.; Zhao, J. The High-Temperature and High-Humidity Storage Behaviors and Electrochemical Degradation Mechanism of LiNi_{0.6}Co_{0.2}Mn_{0.2}O₂ Cathode Material for Lithium Ion Batteries. *J. Power Sources* **2017**, 363, 168–176.
- (15) Renfrew, S. E.; McCloskey, B. D. Residual Lithium Carbonate Predominantly Accounts for First Cycle CO₂ and CO Outgassing of Li-Stoichiometric and Li-Rich Layered Transition-Metal Oxides. *J. Am. Chem. Soc.* **2017**, 139 (49), 17853–17860.
- (16) Liu, X.; Ren, D.; Hsu, H.; Feng, X.; Xu, G.-L.; Zhuang, M.; Gao, H.; Lu, L.; Han, X.; Chu, Z.; Li, J.; He, X.; Amine, K.; Ouyang, M. Thermal Runaway of Lithium-Ion Batteries Without Internal Short Circuit. *Joule* **2018**, 1–19.
- (17) Wang, T.; Maxisch, A.; Ceder, G. A First-Principles Approach to Studying the Thermal Stability of Oxide Cathode Materials. *Chem. Mater.* **2006**, 19 (3), 543–552.
- (18) Zheng, J.; Liu, T.; Hu, Z.; Wei, Y.; Song, X.; Ren, Y.; Wang, W.; Rao, M.; Lin, Y.; Chen, Z.; Lu, J.; Wang, C.; Amine, K.; Pan, F. Tuning of Thermal Stability in Layered Li(Ni_xMn_yCo_z)O₂. *J. Am. Chem. Soc.* **2016**, 138 (40), 13326–13334.
- (19) Wise, A. M.; Ban, C.; Weker, J. N.; Misra, S.; Cavanagh, A. S.; Wu, Z.; Li, Z.; Whittingham, M. S.; Xu, K.; George, S. M.; Toney, M. F. Effect of Al₂O₃ Coating on Stabilizing LiNi_{0.4}Mn_{0.4}Co_{0.2}O₂ Cathodes. *Chem. Mater.* **2015**, 27 (17), 6146–6154.
- (20) Liao, J.-Y.; Manthiram, A. Surface-Modified Concentration-Gradient Ni-Rich Layered Oxide Cathodes for High-Energy Lithium-Ion Batteries. *J. Power Sources* **2015**, 282, 429–436.
- (21) Xie, J.; Sendek, A. D.; Cubuk, E. D.; Zhang, X.; Lu, Z.; Gong, Y.; Wu, T.; Shi, F.; Liu, W.; Reed, E. J.; Cui, Y. Atomic Layer Deposition of Stable LiAlF₄ Lithium Ion Conductive Interfacial Layer for Stable Cathode Cycling. *ACS Nano* **2017**, 11, 7019.
- (22) Tan, K.; Reddy, M.; Rao, G.; Chowdari, B. Effect of AlPO₃ Coating on Cathodic Behaviour of Li(NiCo)O₂. *J. Power Sources* **2005**, 141 (1), 129–142.
- (23) Yan, P.; Zheng, J.; Liu, J.; Wang, B.; Cheng, X.; Zhang, Y.; Sun, X.; Wang, C.; Zhang, J.-G. Tailoring Grain Boundary Structures and Chemistry of Ni-Rich Layered Cathodes for Enhanced Cycle Stability of Lithium-Ion Batteries. *Nature energy* **2018**, 3, 600–605.
- (24) Li, W.; Liu, X.; Celio, H.; Smith, P.; Dolocan, A.; Chi, M.; Manthiram, A. Mn Versus Al in Layered Oxide Cathodes in Lithium-Ion Batteries: a Comprehensive Evaluation on Long-Term Cyclability. *Adv. Energy Mater.* **2018**, 1, 1703154.
- (25) Luo, W.; Li, X.; Dahn, J. R. Synthesis, Characterization, and Thermal Stability of Li[Ni_{1/3}Mn_{1/3}Co_{1/3-z}(MnMg)_{z/2}]O₂. *Chem. Mater.* **2010**, 22 (17), 5065–5073.
- (26) Kam, K. C.; Doeff, M. M. Aliovalent Titanium Substitution in Layered Mixed Li Ni–Mn–Co Oxides for Lithium Battery Applications. *J. Mater. Chem.* **2011**, 21 (27), 9991–9993.
- (27) Tong, D.-G.; Cao, J.-L.; Lai, Q.-Y.; Tang, A.-D.; Huang, K.-L.; He, Y.; Ji, X.-Y. Synthesis and Characterization of LiCo_{0.3-x}Ga_xNi_{0.7}O₂ (x = 0, 0.05) as a Cathode Material for Lithium Ion Battery. *Materials Chemistry and Physics* **2006**, 100 (2–3), 217–223.
- (28) Li, B.; Yan, H.; Ma, J.; Yu, P.; Xia, D.; Huang, W.; Chu, W.; Wu, Z. Manipulating the Electronic Structure of Li-Rich Manganese-Based Oxide Using Polyanions: Towards Better Electrochemical Performance. *Adv. Funct. Mater.* **2014**, 24 (32), 5112–5118.
- (29) Uzun, D. Boron-Doped Li_{1.2}Mn_{0.6}Ni_{0.2}O₂ as a Cathode Active Material for Lithium Ion Battery. *Solid State Ionics* **2015**, 281, 73–81.
- (30) Pan, L.; Xia, Y.; Qiu, B.; Zhao, H.; Guo, H.; Jia, K.; Gu, Q.; Liu, Z. Structure and Electrochemistry of B Doped Li(Li_{0.2}Ni_{0.13}Co_{0.13}Mn_{0.54})_{1-x}B_xO₂ as Cathode Materials for Lithium-Ion Batteries. *J. Power Sources* **2016**, 327, 273–280.
- (31) Han, S.-H.; Song, J. H.; Yim, T.; Kim, Y.-J.; Yu, J.-S.; Yoon, S. Communication-Improvement of Structural Stability During High-Voltage Cycling in High-Nickel Cathode Materials with B₂O₃ Addition. *J. Electrochem. Soc.* **2016**, 163 (5), A748–A750.
- (32) Chen, T.; Li, X.; Wang, H.; Yan, X.; Wang, L.; Deng, B.; Ge, W.; Qu, M. The Effect of Gradient Boracic Polyanion-Doping on Structure, Morphology, and Cycling Performance of Ni-Rich LiNi_{0.8}Co_{0.15}Al_{0.05}O₂ Cathode Material. *J. Power Sources* **2018**, 374, 1–11.
- (33) Yang, X. Q.; Sun, X.; McBreen, J. New Findings on the Phase Transitions in Li_{1-x}NiO₂ in Situ Synchrotron X-Ray Diffraction Studies. *Electrochemistry Communications* **1999**, 1 (6), 227–232.
- (34) Kondrakov, A. O.; Schmidt, A.; Xu, J.; Geßwein, H.; Mönig, R.; Hartmann, P.; Sommer, H.; Brezesinski, T.; Janek, J. Anisotropic Lattice Strain and Mechanical Degradation of High- and Low-Nickel NCM Cathode Materials for Li-Ion Batteries. *J. Phys. Chem. C* **2017**, 121 (6), 3286–3294.
- (35) Ishidzu, K.; Oka, Y.; Nakamura, T. Lattice Volume Change During Charge/Discharge Reaction and Cycle Performance of Li[Ni_xCo_yMn_z]O₂. *Solid State Ionics* **2016**, 288, 176–179.
- (36) A Comparative Study of LiNi_{0.8}Co_{0.2}O₂ Cathode Materials Modified by Lattice-Doping and Surface-Coating. *Solid State Ionics* **2004**, 166 (3–4), 317–325.
- (37) Arie Zaban; Ella Zinigrad, A.; Aurbach, D. Impedance Spectroscopy of Li Electrodes. 4. a General Simple Model of the Li–Solution Interphase in Polar Aprotic Systems. *The Journal of Physical Chemistry* **1996**, 100 (8), 3089–3101.
- (38) Wamser, C. A. Equilibria in the System Boron Trifluoride–Water at 25°. *J. Am. Chem. Soc.* **1951**, 73 (1), 409–416.
- (39) Xu, M.; Zhou, L.; Dong, Y.; Chen, Y.; Demeaux, J.; MacIntosh, A. D.; Garsuch, A.; Lucht, B. L. Development of Novel Lithium Borate Additives for Designed Surface Modification of High Voltage LiNi_{0.5}Mn_{1.5}O₄ Cathodes. *Energy Environ. Sci.* **2016**, 9 (4), 1308–1319.
- (40) Li, Y.; Wan, S.; Veith, G. M.; Unocic, R. R.; Paranthaman, M. P.; Dai, S.; Sun, X. G. A Novel Electrolyte Salt Additive for Lithium-Ion Batteries with Voltages Greater Than 4.7 V. *Adv. Energy Mater.* **2017**, 7 (4), 1601397.
- (41) Li, X.; Zheng, J.; Engelhard, M. H.; Mei, D.; Li, Q.; Jiao, S.; Liu, N.; Zhao, W.; Zhang, J.-G.; Xu, W. Effects of Imide–Orthoborate Dual-Salt Mixtures in Organic Carbonate Electrolytes on the Stability of Lithium Metal Batteries. *ACS Appl. Mater. Interfaces* **2018**, 10 (3), 2469–2479.
- (42) Kim, J.; Ma, H.; Cha, H.; Lee, H.; Sung, J.; Seo, M.; Oh, P.; Park, M.; Cho, J. A Highly Stabilized Nickel-Rich Cathode Material by Nanoscale Epitaxy Control for High-Energy Lithium-Ion Batteries. *Energy Environ. Sci.* **2018**, 11 (6), 1449–1459.
- (43) Ong, C. W.; Huang, H.; Zheng, B.; Kwok, R. W. M.; Hui, Y. Y.; Lau, W. M. X-Ray Photoemission Spectroscopy of Nonmetallic Materials: Electronic Structures of Boron and B_xO_y. *Journal of Applied Physics* **2004**, 95 (7), 3527–3534.
- (44) Oh, P.; Song, B.; Li, W.; Manthiram, A. Overcoming the Chemical Instability on Exposure to Air of Ni-Rich Layered Oxide Cathodes by Coating with Spinel LiMn_{1.9}Al_{0.1}O₄. *Journal of Materials Chemistry A: Materials for energy and sustainability* **2016**, 4 (16), 5839–5841.
- (45) Jain, A.; Hautier, G.; Ong, S. P.; Dacek, S.; Ceder, G. Relating Voltage and Thermal Safety in Li-Ion Battery Cathodes: a High-Throughput Computational Study. *Phys. Chem. Chem. Phys.* **2015**, 17 (8), 5942–5953.
- (46) Ceder, G. Opportunities and Challenges for First-Principles Materials Design and Applications to Li Battery Materials. *MRS Bulletin* **2011**, 35 (09), 693–701.

Table of Contents

